

Two's complement.

1. $(11110111)_2$

$$= -128 + 64 + 32 + 16 + 4 + 2 + 1$$

$$= -128 + 119$$

$$= -9_{10}$$

comp: complement

ver: verification

10 First (-no. $\rightarrow$ binary)
10 sec (binary $\rightarrow$ -no.)
2's complement

Ver:

$$(-9)_{10} = 2's_{comp}(9_{10})$$

$$= 2's_{comp}(00001001)_2$$

$$= 1's_{comp}(00001001)_2 + 1_2$$

$$= \underline{\underline{11110111}}_2$$

$$= 11110110_2 + 1_2$$

2's Complement Problem

$$\textcircled{2} \quad 10111001_2$$

$$= -128 + 32 + 16 + 8 + 1$$

$$= 57 - 128$$

$$= -71$$

Verification

$$-71_{10} = 2's \text{ comp}(01000111)$$

$$= 1's \text{ comp}(01000111_2) + 1_2$$

$$= 10111001_2$$

$$\begin{array}{r} 128 \\ -57 \\ \hline 71 \end{array}$$

$$\begin{array}{r} 2 \overline{) 71} \\ 2 \overline{) 35} - 1 \\ 2 \overline{) 17} - 1 \\ 2 \overline{) 8} - 1 \\ 2 \overline{) 4} - 0 \\ 2 \overline{) 2} - 0 \\ 2 \overline{) 1} - 0 \\ \hline 1 \end{array}$$

$$\textcircled{3} \quad 11110110_2$$

$$= -128 + 64 + 32 + 16 + 4 + 2$$

$$= 128 - 128$$

$$= -10$$

$$\begin{array}{r} 128 \\ - 118 \\ \hline 010 \end{array}$$

Ver:

$$-10_{10} = 2's \text{ comp} (00001010_2)$$

$$= 1's \text{ comp} (00001010_2) + 1_2$$

$$= 11110110_2$$

$$\textcircled{4} \quad 1111101_2$$

$$= (-128 + 64 + 32 + 16 + 8 + 4 + 1)_{10}$$

$$= 125 - 128$$

$$= -3$$

ver:

$$-3_{10} = 2's \text{ comp}(00000011)_2$$

$$= 1's \text{ comp}(00000011)_2 + 1$$

$\therefore$

$=$

$$1111101_2$$

$$\begin{array}{r} 2 \\ 64 \\ + 32 \\ + 16 \\ 8 \\ 4 \\ 1 \\ \hline 125 \end{array}$$

$$\textcircled{5} \quad 10010010_2$$

$$= (-128 + 16 + 2)_{10}$$

$$= 18 - 128$$

$$= -110$$

$$\begin{array}{r} 128 \\ - 18 \\ \hline 110 \end{array}$$

$$11110101$$

$$\begin{array}{r} + \\ \hline 11110110 \end{array}$$

ver:

$$-110_{10}$$

$$= 2's \text{ comp } (01101110)_2$$

$$= 1's \text{ comp } (01101110)_2 + 1_2$$

$$= 10010001$$

$$= 10010010$$

$$2 \overline{) 110}$$

$$2 \overline{) 55-0}$$

$$2 \overline{) 27-1}$$

$$2 \overline{) 13-1}$$

$$2 \overline{) 6-1}$$

$$2 \overline{) 3-0}$$

$$2 \overline{) 1-1}$$

$$2 \overline{) 1}$$

$$\textcircled{6} \ 11010101_2$$

$$= (-128 + 64 + 16 + 4 + 1)_{10}$$

$$= 85 - 128$$

$$= -43$$

$$\begin{array}{r} 64 \\ + 16 \\ + 4 \\ + 1 \\ \hline 85 \end{array}$$

$$\begin{array}{r} 128 \\ - 85 \\ \hline 043 \end{array}$$

ver:

$$-43$$

$$= 2's \text{ comp } (00101011_2)$$

$$= 1's \text{ comp } (00101011_2) + 1_2$$

$$= (11010100 + 1)_2$$

$$= 11010101_2$$

$$\begin{array}{r} 2 \overline{) 43} \\ 2 \overline{) 21} - 1 \\ 2 \overline{) 10} - 1 \\ 2 \overline{) 5} - 0 \\ 2 \overline{) 2} - 1 \\ 2 \overline{) 1} - 0 \\ \hline 1 \end{array}$$

$$\textcircled{7} \ 1111111_2$$

+2+1

$$= (-128 + 64 + 32 + 16 + 8 + 4)_x_{10}$$

$$= -1$$

ver:

$$-1$$

$$= 2's \text{ comp } (0000000)_2$$

$$= 1's \text{ comp } (00000001)_2 + 1_2$$

$$= 11111110_2 + 1_2$$

$$= 11111111$$

$$\begin{array}{r} 2 \\ 64 \end{array}$$

$$+ 32$$

$$+ 16$$

$$+ 8$$

$$+ 4$$

$$+ 2$$

$$+ 1$$

$$\hline 127$$

$$2 \overline{) 127}$$

$$\underline{63-1}$$

$$2 \overline{) 31-1}$$

$$2 \overline{) 15-1}$$

$$2 \overline{) 7-1}$$

$$2 \overline{) 3-1}$$

$$2 \overline{) 1-1}$$

$$1$$

$$\textcircled{8} \ 10000000_2$$

$$= -128 + 0$$

$$= -128$$

ver:

$$-128$$

$$= 2^1s \text{ comp}(128)_{16}$$

$$= 2^1s \text{ comp}(10000000)_2$$

$$= 1^1s \text{ comp}(10000000)_2 + 1_2$$

$$\approx 01111111_2 \times 1_2$$

$$\approx 10000000_2$$

$$\textcircled{9} 11001001_2$$

$$= (-128 + 64 + 8 + 1)$$

$$= 78 - 128$$

$$= -50$$

~~X~~

Do this
20 times

Ver:

-50

$$= 2^1s \text{ comp}(00110010)_2$$

$$= 1^1s \text{ comp}(11001101_2) + 1_2$$

$$= 11001101_2$$

~~X~~

$$\begin{array}{r} -1 \\ 128 \end{array}$$

$$78$$

$$\hline 50$$

$$2 \overline{) 50}$$

$$2 \overline{) 25-0}$$

$$2 \overline{) 12-1}$$

$$2 \overline{) 6-0}$$

$$2 \overline{) 3-0}$$

$$2 \overline{) 1-1}$$

1

$$\textcircled{10} \ 11100111_2$$

$$= (-128 + 64 + 32 + 4 + 2 + 1)$$

$$= 103 - 128$$

$$= -25$$

$$\begin{array}{r} 128 \\ -103 \\ \hline 025 \end{array}$$

ver:

$$-25$$

$$= 2's \text{ comp } (00011001_2)$$

$$= 1's \text{ comp } (00011001_2) + 1_2$$

$$= 11100111_2$$

$$\begin{array}{r} 2 \overline{) 25} \\ 2 \overline{) 12-1} \\ 2 \overline{) 6-0} \\ 2 \overline{) 3-0} \\ 2 \overline{) 1-1} \\ 1 \end{array}$$

(ii) -20_{10}

$$20_{10} = (00010100)_2$$

$$1's \text{ comp: } (11101011)_2$$

2's comp:

$$11101100$$

Ver:

$$= -128 + 64 + 32 + 0 + 8 + 4 + 0 + 0$$

$$= -20_{10}$$

$$\begin{array}{r} + 1011 \\ \hline 1100 \end{array}$$

Step by step:

$$\begin{aligned} -20_{10} &= \text{Two's Complement}(20_{10}) \\ &= \text{One's Complement}(20_{10}) + 1 \\ &= \text{One's Complement}(00010100)_2 + 1 \end{aligned}$$

$$= 1101011_2 + 1$$

$$= 1101100_2$$

Ver: $11101100_2 = -128 + 64 + 32 + 8 + 4 = -20$

$$\textcircled{12} - 96$$

$$= 2's \text{comp}(01100000_2)$$

$$= 1's \text{comp}(10011111_2) + 1_2$$

$$= 10100000_2$$

$$\begin{array}{r} 2 \overline{) 96} \\ \underline{2 48} \\ 2 24 \\ \underline{2 12} \\ 2 6 \\ \underline{2 3} \\ 2 1 \\ \underline{ 0} \end{array}$$

$$10100000_2$$

ver:

$$= -128 + 32$$

$$= -96$$

$$\begin{array}{r} -1 \\ 128 \\ - 32 \\ \hline 096 \end{array}$$

$$\textcircled{13} - 21$$

$$- 21_{10}$$

$$= 2's_{comp}(11101010)_2$$

$$= 1's_{comp}(11101010)_2 + 1_2$$

$$= (11101011)_2$$

Verification

$$(11101011)_2$$

$$= -128 + 64 + 32 + 8 + 2 + 1$$

$$= -21_{10}$$

$$\textcircled{14} -16$$

$$= 2's \text{ comp}(00010000_2)$$

$$= 1's \text{ comp}(11101111) + 1_2$$

$$= 1110000_2$$

$$\begin{array}{r} 2 \overline{) 16} \\ 2 \overline{) 8 - 0} \\ 2 \overline{) 4 - 0} \\ 2 \overline{) 2 - 0} \\ 2 \overline{) 1 - 0} \\ \hline 1 \end{array}$$

ver:

$$1110000 = -128 + 64 + 32 + 16$$

$$= -16$$

15 -9

$$\begin{aligned} 2's \text{ comp}(-9_{10}) &= 2's \text{ comp}(00001001_2) \\ &= 1's \text{ comp}(11110110_2) + 1 \\ &= 11110111_2 \end{aligned}$$

11110111₂

ver:

$$= -(2^8 + 2^6 + 2^5 + 2^4 + 2^2 + 2^1)$$

$$= -9$$

$$\begin{array}{r} 11110110 \\ + 1 \\ \hline 11110111 \end{array}$$

$$\begin{array}{r} 128 \\ - 119 \\ \hline 009 \end{array}$$

$$\begin{array}{r} 64 \\ 32 \\ 16 \\ 4 \\ 2 \\ 1 \\ \hline 119 \end{array}$$

$$\textcircled{16} -36_{10}$$

$$2's \text{ comp of } (-36_{10})$$

$$= 2's \text{ comp of } (00100100_2)$$

$$= 1's \text{ comp of } (00100100_2) + 1$$

$$= 11011100_2$$

$$2 \overline{) 36}$$

$$2 \overline{) 18 - 0}$$

$$2 \overline{) 9 - 0}$$

$$2 \overline{) 4 - 1}$$

$$2 \overline{) 2 - 0}$$

$$2 \overline{) 1 - 0}$$

$$\begin{array}{r} 1 \\ 128 \end{array}$$

$$\begin{array}{r} - 92 \\ \hline 036 \end{array}$$

$$1101100_2 = \overset{\text{ver}}{-128} + 64 + 16 + 8 + 4$$

$$= -36$$

$$\begin{array}{r} 11011011 \\ + \\ \hline 11011100 \end{array}$$

$$\textcircled{17} -50$$

$$= 2's \text{ comp}(00110010)_2$$

$$= 1's \text{ comp}(00110010)_2 + 1_2$$

$$= 11001101 + 1_2 = 11001110_2$$

↓ ↓ ↓

32 16

$$-1-50 = -51$$

$$= 11001110_2$$

ver:

11001110

$$= -128 + 64 + 8 + 4 + 2 = -50$$

$$2 \overline{) 50}$$

$$2 \overline{) 25-0}$$

$$2 \overline{) 12-1}$$

$$2 \overline{) 6-0}$$

$$2 \overline{) 3-0}$$

$$2 \overline{) 1-1}$$

$$+ \begin{array}{r} 64 \\ 8 \\ 4 \\ 2 \\ \hline 78 \end{array}$$

$$\begin{array}{r} 128 \\ - 78 \\ \hline 50 \end{array}$$

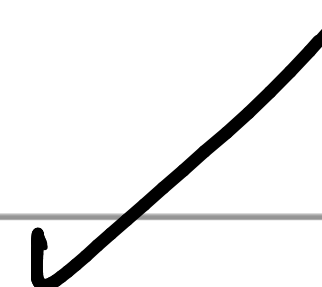
$$\textcircled{18} -101$$

$$= 2's \text{ comp}(01100101_2)$$

$$= 1's \text{ comp}(01100101_2) + 1_2$$

$$= 10011010_2 + 1_2$$

$$= 10011011$$

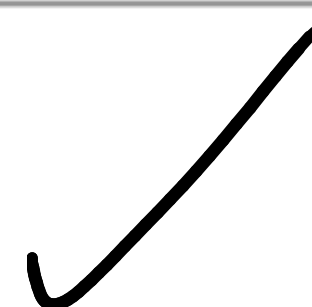


ver:

$$10011011$$

$$= -128 + 16 + 8 + 2 + 1$$

$$= -101$$



$$2 \overline{) 101}$$

$$2 \overline{) 50-1}$$

$$2 \overline{) 25-0}$$

$$2 \overline{) 12-1}$$

$$2 \overline{) 6-0}$$

$$2 \overline{) 3-0}$$

$$2 \overline{) 1-1}$$

1

$$\textcircled{19} - 123_{10}$$

$$= 2's \text{ comp}(0111011_2)$$

$$= 1's \text{ comp}(0111011_2) + 1_2$$

$$= 10000100_2 + 1_2$$

$$= 10000101$$

10000101 ver:

$$= -128 + 4 + 1$$

$$= -128 + 5$$

$$= -123$$

$$\begin{array}{r} 2 \overline{) 123} \\ \underline{6} \\ 2 \overline{) 61} \\ \underline{4} \\ 2 \overline{) 20} \\ \underline{15} \\ 2 \overline{) 51} \\ \underline{4} \\ 2 \overline{) 31} \\ \underline{2} \\ 2 \overline{) 11} \\ \underline{1} \\ 1 \end{array}$$

$$\textcircled{20} - 5$$

$$= 2's \text{ comp } (00000101)_2$$

$$= 1's \text{ comp}$$

$$(00000101)_2 + 1$$

$$= 1111010 + 1$$

$$= 1111011$$

$$\begin{array}{r}
 2 \\
 64 \\
 32 \\
 + 16 \\
 8 \\
 2 \\
 1 \\
 \hline
 123 \\
 - 128 \\
 \hline
 123 \\
 \hline
 005
 \end{array}$$

$$1111011$$

ver:

$$= -128 + 64 + 32 + 16 + 8 + 2 + 1$$

$$= -5$$